

Analysing the efficacy of UML in explaining object-oriented concepts to undergraduate computer science students

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Abstract—This study used the undergraduate course Object-Oriented Analysis and Design as a case study to assess how the depth of knowledge of object-oriented concepts was developed in preparation for learning the Unified Modeling Language (UML). Two surveys were conducted to assess student learning. The first survey identified the concepts of object orientation that students should be aware of while taking object-oriented courses. The second survey examined the depth of knowledge gained from learning UML. The results show that teaching UML significantly increases students' knowledge of UML compared to just learning it in object-oriented courses. This study concluded with reflections on the effectiveness of UML as an approach to developing the concept when taught at the undergraduate level.

Index Terms—Object-oriented concept, UML, Statistical data, Qualitative analysis

I. INTRODUCTION

UML stands for Unified Modeling Language and is very popular for modeling business systems and helps in the analysis, design, and implementation of any system. It combines the best diagramming practices used by software developers or analysts. It gives the software developer the freedom to design the system according to his own preferences and techniques while ensuring that he understands the requirements properly. Object-oriented programming is a challenging task for undergraduate programs, as it extends to all tertiary educational institutions. Several courses are offered in undergraduate computer science programs that cover OOP concepts as well as languages such as Java, and C++ [1], [2]. In the undergraduate

computer science curriculum, UML is taught in the Object-Oriented Analysis and Design (OOAD) course [3]. Philip and Bruhn explained the effectiveness of UML in teaching the OOAD course. By using UML, students are actively engaged in the learning process, and it increases their interest in creating a more complete picture of a problem through diagrams [4], [5].

As in other universities, the undergraduate computer science program (CS) of American International University-Bangladesh (AIUB) covers both UML and object-oriented concepts. Students in the undergraduate CS program take several courses based on the object-oriented concept. In the Department of Computer Science curriculum, there is a course titled "Object-Oriented Analysis and Design" that covers details of UML [3]. In this research, we aimed to find out if the use of UML has an impact on the understanding of the object-oriented concept. Therefore, this work mainly focuses on the understanding of the object-oriented concept among undergraduate students who use UML.

In this article we evaluated how UML improves the understanding of the object-oriented concept. The empirical study addresses the following research questions:

- 1) **RQ1:** Do students who do not know UML have a good understanding of object orientation?
- 2) **RQ2:** Does knowledge of UML have an impact on understanding of object-oriented language?

To answer the RQs, we selected the undergraduate course Object-Oriented Analysis and Design in Computer Science. The study included two surveys to assess student learning. The first was conducted to determine the learning of object-oriented concepts in various object-oriented courses. The second survey examined the depth of knowledge of object orientation, taking into account UML learning. In both surveys, different categories of students were recorded according to their academic results and course completion. Finally, the survey results showed that learning UML significantly improved students' understanding.

The remainder of the article is organized as follows. Section II provides the related study, an overview of the survey procedures, and respondent selection. The outcomes of the survey results are presented in section III. Section IV discusses the survey results and explains the results' validity using statistical rules. Finally, section six concludes the paper with an outlook on the future scope of the research conducted.

II. MATERIALS AND METHODS

A. Related Study

The related study focuses primarily on the relationship and use of the object-oriented concept and UML in tertiary-level education and industry. Data were collected from various sources such as journals, conference proceedings, and books on UML and object-oriented concepts. All relevant data were collected and analyzed to gather relevant information.

Unified Modeling Language (UML) is a notation technique that represents the models used to design an application or business model. UML is one of the most widely used modeling languages in the software industry because it simplifies the design of complex processes with graphical notations [6]. Data from GitHub shows that UML enjoys high adoption in the software industry, with a large number of projects using UML modeling [7]. In object-oriented analysis, the most important aspects of a system are worked out, i.e., what the system does and how it is being done. On the other hand, how the system performs its functionalities is covered by the object-oriented design [8], [9]. The systems analyst uses UML to analyze and design a business process. UML uses object modeling notations to represent the interactions of objects and the collaboration between classes. UML gives developers much freedom because they can use their own style and choice to design a system considering the actual functionalities of the product or system [8]–[10].

Several tools have been implemented and are being used to help undergraduate students learn object-oriented design with UML. For learning UML quickly, Umple is a very efficient tool [11]. With Umple, students can successfully model their assignments, exercises, and projects, which is reflected in their grades [12]. Considering the teaching experience of the OOAD course, a tool called StudentUML [13] supports the learning process in both undergraduate and postgraduate education. The tool enriches the software development process of students by ensuring correctness and consistency in modeling [14]. Some of the challenges and opportunities in modeling education

were identified by Leibel et al. through a case study of model-driven software development. Several factors influence the use of a tool for students to understand the effectiveness of modeling [15], [16].

The UML model examples have increased student confidence in model construction and also increased the quality of solutions. UML is the vehicle for teaching core concepts from software engineering, and some of the fundamental principles of computer science in general [17]. It also improves understanding of collaboration between classes, the different roles of elements, and the structure of the class diagram [18]. Teaching UML to students using the notations for modeling will spur a career, and the best teaching concepts can fail without adequate operationalization performance. UML is the practical approach to correctly understanding object orientation and abstraction and is able to clarify the core concepts of software engineering [17], [19].

By examining the above industry solutions and practices, we find out what impact UML has on the tertiary-level of education. However, the results of our study ensure that UML has an impact on improving the object-oriented concept. The results also describe the validity of the results obtained by different students.

B. Survey Overview

The goal of the surveys was to examine students' understanding of the object-oriented concept. Students can learn this concept in different courses based on the object-oriented concept. The study was qualitative in nature and was divided into two assessments. These were a pre-assessment and a post-assessment. Both assessments contained several qualitative questions, and we ensured that the same set of questions was used in both the pre-assessment and post-assessment to maintain the validity of the comparison.

For the pre-assessment, we selected students who had at least completed the courses on object-oriented concepts such as C++, Java, etc. The primary purpose of the pre-assessment is to find out if the students have any idea about object orientation. Also, we try to find out whether the students are able to represent the real-world scenarios with objects and classes or not.

For the post-assessment, the students who had completed the object-oriented programming courses and Object-Oriented Analysis and Design (OOAD) courses were selected. In the OOAD course, students learn about UML. The post-assessment mainly focuses on proper reasoning about object orientation according to the principles and designing a real-world problem through objects and classes.

C. Participants

Surveys 1 and 2 were assessed in two different academic cycles. In the first cycle (3rd/4th-semester), students had completed the object-oriented concept courses but knew nothing about UML. And the second cycle (7th/8th-semester students) included the students who learned the idea of UML from the OOAD course [20]. Basically, the program AIUB CS

includes several courses on object-oriented concepts [3]. In addition to these object-oriented programming courses, the OOAD course is also taught at the undergraduate level. It is a perfect approach to familiarize students with UML while they learn the concept of object orientation in various programming language courses. According to the AIUB curriculum, students get the knowledge of object orientation before learning UML.

This study refers to the students who participated in the survey and were enrolled in the undergraduate Computer Science (CS) program at AIUB. All of the participating students were familiar with the concept of object orientation because they had already taken various courses that followed the object-oriented concept. However, about 50% of them had only taken the "Object-Oriented Analysis and Design" course, which basically covers the details of UML. The students who only knew the concept of object orientation were selected for the preliminary investigation.

D. Survey Procedure and data analysis

Both surveys were self-administered surveys in which students were given an explanation of the goal of the survey. Students were required to answer all questions, as all questions were mandatory. The survey did not collect any personally identifiable information from participants with a view to later publishing the survey results.

In the first cycle, 3rd/4th-semester students participated in the pre-assessment process. Seventy students submitted the first survey questionnaire responses. For the second cycle, 7th/8th-semester students were contributed for post-assessment, and the number of contributors was fifty-four who had submitted the responses. In both cycles, the first and second respondents completed the surveys by answering questions in sequential order. Table I shows the total number of participants' data.

TABLE I
PARTICIPANT RESPONDENTS FOR SURVEY 1 AND 2

Survey	No of Participants	Details of Survey Categories
Survey 1 (unknown to UML)	n= 70	Pre-assessment Survey, first cycle (3rd/4th-semester students)
Survey 2 (Knows about UML)	n= 54	Post-assessment Survey, second cycle (7th/8th-semester students)

To examine the impact of UML on student learning, statistical data sets from Survey 1 and Survey 2 were analyzed. Surveys 1 and 2 included fifteen individual questions whose responses assessed knowledge of object orientations. Using the survey dataset, students were categorized based on the correct answer. To avoid bias in the results, students were also categorized based on their academic performance.

III. RESULTS

All categories of students participated in the survey; therefore, we also distinguished the participants according to their grade point average. We divided the resulting scores into three categories based on the percentage of the correct answer to the questionnaire. If the correct answer was more than 66 percent,

it was classified as high, below 33 percent as low, and in between as average, as described in Table II. This scoring method was used in both surveys.

TABLE II
CORRECT ANSWER RESPONDENTS FROM SURVEY 1 AND 2

Classification	Name	Criteria
Category-1	High	if the value is greater than 66%
Category-2	Average	if the value lies between 33% to 66%
Category-3	Low	if the value is less than 33%

A. Pre-assessment (survey 1)

In the preliminary evaluation, we found that the highest CGPA was 3.97, the lowest was 2.42, and the average was 3.15. The CGPA was calculated using the AIUB grading system, [21]. With regard to the CGPA, we divided the participants into three different groups. We assigned a CGPA range to all individual groups. Using the CGPA range, we determined the percentage of correct answers in each group.

Using the cumulative grade point average (CGPA) of all participants, we determined different types of results. We found that of the 70 participants in Survey 1, 27 had a high-grade point average, 29 had an average grade point average, and the remaining 14 had a low-grade point average. Judging by the number of participants in the different categories, the high and average ranges accounted for nearly 80% of the total, and the low range accounted for only 20%. Although the low range was almost half the high and average range, this contribution still ensures that all categories of participants took part in Survey 1. Figure 1 illustrates the data for the various participants.

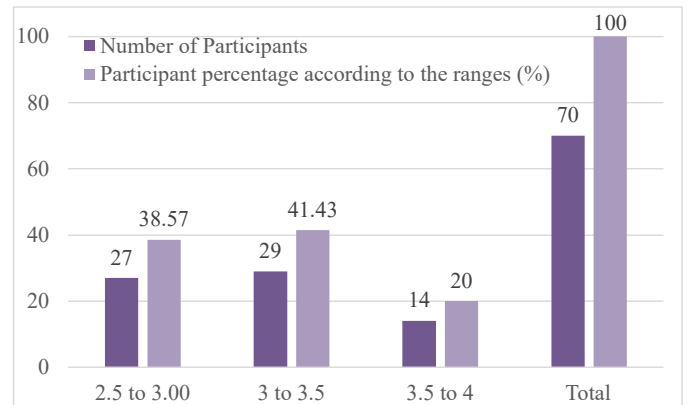


Fig. 1. CGPA range wise participants classification for all respondents from survey 1

Looking at the correct answers from the survey 1 questionnaires, we found that 45.52% of the answers were correct. The percentage of incorrect answers was 54.48. So, these data prove that the participants have acquired basic knowledge but not a deep understanding of object-orientation. The results of the first survey are shown in Table III.

From the corrected answers of the preliminary evaluation, we formulated a standard evaluation criterion. Basically, we

TABLE III
ALL PARTICIPANT AVERAGE RESULTS FOR SURVEY 1

No of student participation	70
Average of the correct answer	45.52%
Average of the incorrect answer	54.48%

divided the average response correction of respondents into three categories, which are shown in Figure 2. If the average percentage is below 33, it is a low group; if the average is above 66, it is a high group; and between 34 and 65 is the middle group. Thus, according to these measures, we found that the middle category contains about 79%, which is the maximum, and that 55 out of 70 students fall into this group. The high group contains about 11% and the low group only 10%, which is very little compared to the medium group.

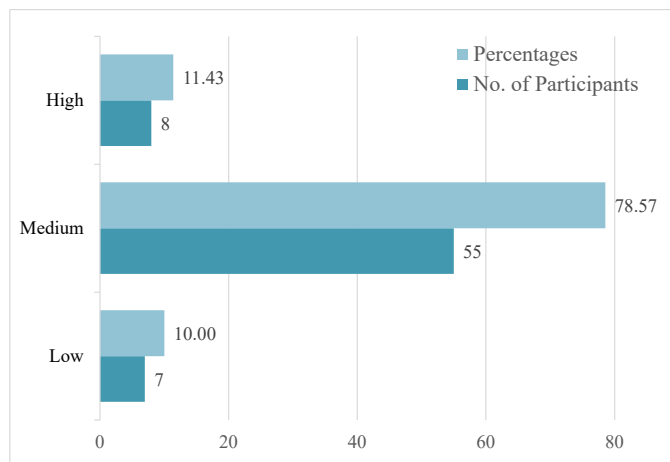


Fig. 2. Group wise participants average results for survey 1

B. Post-assessment (survey 2)

In the post-assessment, we found that the highest CGPA was 3.91, the lowest was 2.62, and the average was 3.14. Following the pre-assessment, we applied the same procedure of CGPA bandwidth clustering in the post-assessment.

In Survey 2, 54 students participated. Among these participants, 24 students were classified into the high CGPA category based on their academic performance. For the average category, there were 18, and for the low category, there were 12. According to the categories, the maximum number of students belongs to the high category. So, we found that the percentage for the high, average, and low categories were 45%, 33%, and 22%, respectively. The percentage breakdown by group ensures that all students participate in the survey, as can be seen in Figure 3.

From the survey 2 questionnaires, we found that the average corrected response was 51%, and the percentage for an incorrect response was 49%, as described in Table IV. With respect to Survey 1, we found that the average correctness of responses to survey questions had increased. The difference between Survey 1 and Survey 2 was 6%. These data demonstrate that

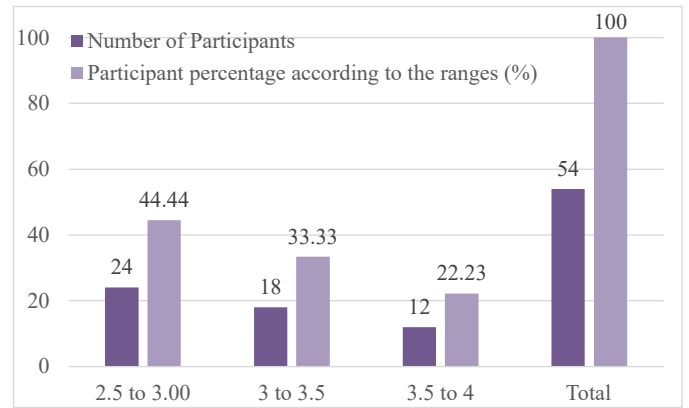


Fig. 3. CGPA range wise participants classification for all respondents from survey 2

in the second cycle, students improved their knowledge with the help of learning UML.

TABLE IV
ALL PARTICIPANTS AVERAGE RESULTS FOR SURVEY 2

No of student participation	54
Average of the correct answer	51.23%
Average of the incorrect answer	48.77%

Based on the average correct response of the respondents, we classified them into the same categories that we had assessed in Survey 1. According to this criterion, we found that the low category, with about 9%, was the minimum value, and 5 out of 54 students fell into this group. The medium category had the highest value of 65%. On the other hand, the high category has a value of 26%, which is increased compared to survey 1. The increase in the value of the high category in Survey 2 reflects that the understanding of object orientation has improved and is shown in Figure 4.

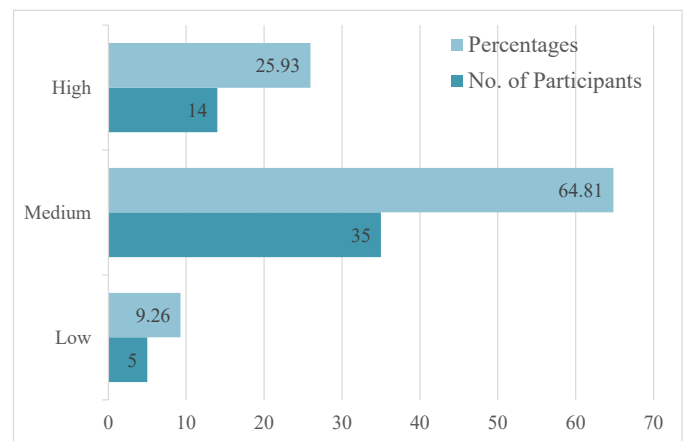


Fig. 4. Group wise participants average results for survey 2

IV. DISCUSSION

This is a qualitative analysis of case studies among students in the Object-Oriented Analysis and Design course.

Both assessments contained multiple questions and were both qualitative in nature. The comparison scopes are,

- The two assessments are significantly different from each other from Z calculation.
- The two assessments have achieved different average values.
- The value of the High category for both assessments is proportionally different.
- The percentage of the High category has been changed from pre to post assessment.
- The value of Low category for both assessments is proportionally different.
- The percentage of the Low category has been changed from pre to post assessment.

Since the student evaluation questionnaires were completed at different levels, we need to find out whether the evaluations are different or not. To prove this, we calculated the Z to find out if the assessments differed. From the post and pre-assessments, we had some values, such as mean, variance, number of participants, etc. From these values, we calculated the Z .

$$Z = (\bar{x}_1 - \bar{x}_2) / \sqrt{s^2(1/n_1 + 1/n_2)}$$

$$= (51.23 - 45.52) / \sqrt{202.85(1/70 + 1/54)}$$

$$= 2.93$$

Where,

$$\bar{x}_1 = 51.23$$

$$\bar{x}_2 = 45.52$$

$$s_1 = 204.32$$

$$s_2 = 200.93$$

$$s = (n_1 - 1)s_1^2 + (n_2 - 1)s_2^2 / n_1 + n_2 - 2$$

$$= 202.85$$

Note: $\bar{x}$ is the mean and s is the variance

As $Z > 1.96$, the two assessments are significantly different. From the Z calculation, we found that the Pre-assessment and the Post-Assessment are significantly different from the calculated value for Z is over 1.96.

$$Z = |(p_1 - p_2) / \sqrt{p * q(1/n_1 + 1/n_2)}|$$

$$= |(0.11 - 0.26) / \sqrt{0.18 * 0.82(1/70 + 1/54)}|$$

$$= |-0.15/0.065|$$

$$= 2.31$$

Where,

$$p_1 = 8/70 = 0.11$$

$$p_2 = 14/54 = 0.26$$

$$p = 8 + 14/70 + 54 = 0.18$$

$$q = 1 - p = 1 - 0.18 = 0.82$$

As $Z > 1.96$, the value of the High category for both assessments is proportionally different.

$$Z = |(p_1 - p_2) / \sqrt{p * q(1/n_1 + 1/n_2)}|$$

$$= |(0.1 - 0.09) / \sqrt{0.09 * 0.91(1/70 + 1/54)}|$$

$$= |0.01/0.0027|$$

$$= 3.7$$

Where,

$$p_1 = 7/70 = 0.1$$

$$p_2 = 5/54 = 0.09$$

$$p = 7 + 5/70 + 54 = 0.09$$

$$q = 1 - p = 1 - 0.09 = 0.91$$

As $Z > 1.96$, the value of Low category for both assessments is proportionally different.

For both assessments, we had three individual categories corresponding to the average response correction of the respondents, which is shown in Figure 5. When we compare Survey 1 with Survey 2, this categorization yields the following,

- In terms of "High" category, the percentage, as well as the no of students, had been increased.
- On the other hand, for "Low" category the percentage and also the no of students had been decreased.

We can conclude the survey result with the following findings,

- From the Pre-assessments, we have found, 11.43% belongs to a High category, 78.57% to Medium category and 10% to the Low category.
- From the Post-assessments we have found, 25.93% belongs to a High category, 64.81% to Medium category and 9.26% to the Low category.
- From the survey result, it seems that two surveys are significantly different from the Z calculation value and on the other hand, for individual category analysis using proportion we have found that both of them were proportionally different.
- Finally, from the survey result, we have found the effectiveness of the UML as the average had been increased from 45.52% to 51.23%. Figure 5 illustrates that the average for post-assessment is much greater than pre-assessment.

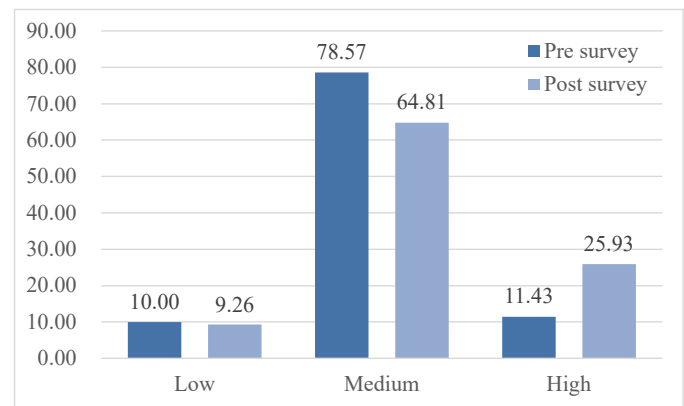


Fig. 5. Post and Pre assessments data in different categories

The results of our study show that UML has a good impact on the understanding of object-orientation. The quantitative results indicate interesting learning outcomes with respect to the two groups of different sessions of students. The survey data reflects that students who do not know about UML

have a lower benefit compared to those who do know about UML. Our study results suggest that UML strongly influenced students' learning of object-oriented languages and triggered positive changes in their knowledge.

V. CONCLUSION

Among the selected survey participants, this research found relevant evidence that UML has an impact on the understanding of the object-oriented concept. Examination of the survey result shows that the two surveys are significantly different from the Z-score of the calculation. The score has increased from 45.52% to 51.23%. All categories are proportionally different, which was proven by the proportional Z calculation, as the value is greater than 1.96. The understanding of the object-oriented concept has increased in the high category, varying from 11.43% to 25.93%. The value for the low category decreased from 10% to 9.26%. Moreover, the understanding of the object-oriented concept increased in the high category, and the value decreased in the low category. Thus, the surveys have successfully shown that the understanding has improved by learning UML.

It seems that students in the undergraduate CS program generally took several courses in a semester. If a student has taken a course on object-oriented analysis and design and object-oriented programming simultaneously in a semester, we can find out the impact of UML in real-time based on these students. In addition, cross-organizational survey data will be very helpful in making an efficient decision. Our study focused on only one university. So if we collect data from different universities, we can make a more specific and accurate decision which is the future direction of this work.

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